

Automatic GFL/GFM Reference-Mode Switching in a 1-SG / 4-IBR IEEE-14 System: Model, Small-Signal Spectrum, and Switched Time-Domain Formulation

Power-flow / stability toolchain — generated report

Abstract

A modified IEEE-14 network is driven through a six-period chronology in which a single synchronous generator (SG) is tripped, the load is stepped, a shunt fault is applied and cleared, a tie line is opened, and the SG is finally reclosed. Four inverter-based resources (IBRs) switch *automatically* between grid-following (GFL) and grid-forming (GFM) reference behaviour under an amplitude–frequency severity index with hysteresis and dwell. This report documents, from the production code, (i) the severity index and its two component sub-indices J_V, J_f ; (ii) the sixth-order SG model with an order-by-order derivation and the singular limit that freezes one flux state; (iii) the twenty-state dual-mode IBR superset with its active projections and current-continuity-gated transfer; (iv) the switched implicit-trapezoidal time-domain formulation, including the exact mechanism by which an integrated state update is replaced by a discrete transfer at a mode change; and (v) the solved power flow, the small-signal spectrum, and the accepted transient response. All equations are those the 250-s run executes; all plotted curves are raw accepted samples of the preserved 200-s trajectory. No claim of operational readiness is made: convergence of the simulation is not evidence of grid-code compliance.

1 Scope, sources and reproduction

The study system is the IEEE-14 bus network with one modelled SG at the reference bus and four modelled IBRs at buses 2, 3, 6 and 8. The whole system is assembled as a single all-KCL composite differential–algebraic system

$$g(x, V) = Y_{\text{net}} V - I_{\text{bus}}(x, V) = 0, \quad (1)$$

where Y_{net} is the network admittance, V the stacked complex bus voltages, and I_{bus} the total device injection. The SG and every IBR contribute their own differential state block x and their own injection into (1); no device is delegated to an external solver, model or reference program. Audited linear primitives (`\`, `lu`, `qr`, `eig`) are the only numerical libraries used.

Equation source. Every model equation in Sections 3–6 is transcribed verbatim from the production code that the chronology driver executes. The driver, its event schedule and its supervisor gains are fixed by

```
s = cases.scenario.ieee14.1sg.4ibr(struct('case_profile',
'eecon49.figure4'));
run.ieee14.gfl.gfm.state.release % t_end=250, dt=0.1
```

The case profile name above is a code identifier and is shown only in this reproduction command.

Figure source and no-synthetic-signal contract. The transient figures (Section 9) are rendered from the preserved accepted trajectory output/diagnostics/engine_release_200s_preserved.mat. This is a deliberate choice: the equations are taken from the current 250-s code, while the plotted curves are the established 200-s results. Every plotted sample is a raw accepted value. Nothing is smoothed, filtered, decimated, clipped, offset, interpolated or augmented with synthetic noise; the figure generator records `presentation_noise=struct('kind','none','seed_base',NaN,'affects_solver_or_switching',false)`. Where a display window is applied it is a horizontal range only — no in-window sample is altered or dropped.

Classification. The SG AC/flux equations and the IBR AC/control equations are SOURCE_MAPPED. The fixed 20-state IBR superset, the GFL↔GFM transfer map and the DC-source regulator closure are PROJECT_DERIVED. Machine and converter parameters, per-unit bases and the disturbance times are CASE_DEFINED. The predictor and step-subdivision policy are NUMERICAL_METHOD. These labels are carried from the code provenance fields and are not re-decided here.

2 Disturbance chronology

The scheduled disturbance times are CASE_DEFINED constants of the event structure; the reclose and severity-release instants are outcomes read from the accepted event log. Figure 1 shows the six operating periods and the transition instants. The shunt fault is applied at $t = 85.000$ s and cleared at $t = 85.150$ s (a 0.150 s bus-9 fault, impedance $Z_f = 0.01 + j0.01$ p.u.); this exact window is taken from the driver and supersedes any wider nominal fault interval.

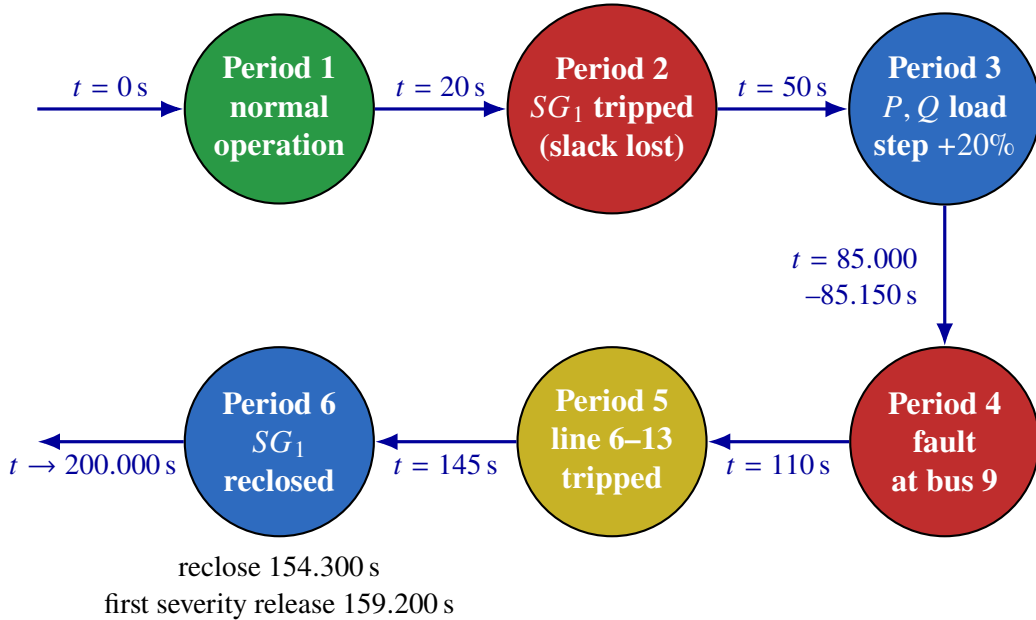


Figure 1: Six-period disturbance chronology. Green: undisturbed. Red: a contingency that removes a source or faults a bus. Yellow: a topology change. Blue: a restorative action. Scheduled instants (0, 20, 50, 85.000–85.150, 110, 145 s) are case-defined; the reclose and first severity-release instants are read from the accepted event log.

3 Automatic GFL/GFM selection: the severity index

Each IBR carries a two-state supervisor that decides whether it should behave as a grid-following current source (GFL) or as a grid-forming voltage source (GFM). The decision is driven by a scalar severity $S(t) \in [0, 1]$ built from two normalised sub-indices: a voltage-deviation index J_V and a frequency-deviation index J_f .

Sub-indices. Let $|V_i(t)|$ be the terminal voltage magnitude of IBR i and V_i^* the healthy per-bus power-flow magnitude at the SG-online operating point (Section 7). Let $f_{\text{coi}}(t)$ be the centre-of-inertia frequency. With bases $\Delta V_{\text{base}} = 0.10$ p.u. and $\Delta f_{\text{base}} = 0.50$ Hz,

$$J_V(t) = \frac{||V_i(t)| - V_i^*|}{\Delta V_{\text{base}}}, \quad J_f(t) = \frac{|f_{\text{coi}}(t) - f_0|}{\Delta f_{\text{base}}}, \quad f_0 = 60 \text{ Hz}. \quad (2)$$

Two-term severity (AGSI). The severity is the saturated equal-weight average

$$S(t) = \text{sat}_{[0,1]} \left(\frac{1}{2} J_V(t) + \frac{1}{2} J_f(t) \right), \quad \text{sat}_{[0,1]}(z) = \min\{1, \max\{0, z\}\}. \quad (3)$$

J_V responds to a local voltage collapse or overshoot; J_f responds to a system-wide frequency excursion such as the one that follows the loss of the SG slack at $t = 20$ s. The equal weighting is a PROJECT_DERIVED choice and is frozen before any result is read.

Hysteresis and dwell. A bare threshold on S would chatter. The supervisor therefore uses an asymmetric two-level comparator with minimum dwell times:

$$\begin{aligned} \text{GFL} \rightarrow \text{GFM} : \quad S &\geq \Gamma_{\text{on}} = 0.65 \text{ held for } T_{d,\text{on}} = 0.10 \text{ s}, \\ \text{GFM} \rightarrow \text{GFL} : \quad S &\leq \Gamma_{\text{off}} = 0.35 \text{ held for } T_{d,\text{off}} = 1.00 \text{ s}. \end{aligned} \quad (4)$$

The gap $\Gamma_{\text{on}} > \Gamma_{\text{off}}$ gives the comparator its hysteresis band; the asymmetric dwell ($T_{d,\text{on}} \ll T_{d,\text{off}}$) makes the supervisor quick to add grid-forming support and deliberately slow to release it. A separate reference-loss guard forces every online IBR to GFM the instant the SG slack is lost, before the dwell logic runs, so the network never spends a step without a voltage reference.

4 Synchronous generator: sixth-order model and its proof

The SG is a sixth-order machine (IEEE 1110 Model 2.2 / round-rotor GENTPJ structure) with state

$$x_{\text{sg}} = [\delta, \omega, E'_q, E'_d, E''_q, E''_d]^\top, \quad u_{\text{sg}} = [T_m, E_{fd}]^\top, \quad (5)$$

where δ is the rotor angle, ω the per-unit speed deviation (slip), E'_q, E'_d the transient EMFs, and E''_q, E''_d the subtransient EMFs. The two control inputs T_m (mechanical torque) and E_{fd} (field voltage) are held at the constant values determined by the mixed-network equilibrium solve; they are not re-solved each step.

4.1 Order-by-order derivation

The connected-machine equations are, term for term as coded,

$$\dot{\delta} = \omega_0 \omega, \quad (6)$$

$$2H \dot{\omega} = T_m - T_e - D \omega, \quad (7)$$

$$T'_{d0} \dot{E}'_q = E_{fd} + c_d E''_q - d_d E'_q, \quad (8)$$

$$T'_{q0} \dot{E}'_d = c_q E''_d - d_q E'_d, \quad (9)$$

$$T''_{d0} \dot{E}''_q = E'_q - E''_q - (X'_d - X''_d) I_d, \quad (10)$$

$$T''_{q0} \dot{E}''_d = E'_d - E''_d + (X'_q - X''_q) I_q, \quad (11)$$

with $\omega_0 = 2\pi f_0$. The air-gap electrical torque that closes the swing equation (7) is

$$T_e = V_d I_d + V_q I_q + R_a (I_d^2 + I_q^2), \quad (12)$$

and the four flux-coupling coefficients are defined from the reactance ladder as

$$c_d = \frac{X_d - X'_d}{X'_d - X''_d}, \quad d_d = \frac{X_d - X''_d}{X'_d - X''_d}, \quad c_q = \frac{X_q - X'_q}{X'_q - X''_q}, \quad d_q = \frac{X_q - X''_q}{X'_q - X''_q}. \quad (13)$$

What each order is. Equations (6)–(7) are the mechanical (2nd-order) swing: angle and speed. Equations (8)–(9) are the transient (rotor field / q-axis damper) fluxes on the slow time constants T'_{d0}, T'_{q0} . Equations (10)–(11) are the subtransient (damper) fluxes on the fast time constants T''_{d0}, T''_{q0} . The armature-reaction terms $-(X'_d - X''_d)I_d$ and $+(X'_q - X''_q)I_q$ couple the fast fluxes to the stator current.

Proof anchor — the $d - c = 1$ identity. From (13),

$$d_d - c_d = \frac{(X_d - X''_d) - (X_d - X'_d)}{X'_d - X''_d} = \frac{X'_d - X''_d}{X'_d - X''_d} = 1, \quad \text{and identically } d_q - c_q = 1. \quad (14)$$

Consider the open-circuit steady state ($I_d = I_q = 0, \dot{x} = 0$). Then (10) gives $E''_q = E'_q$, and substituting into (8),

$$0 = E_{fd} + c_d E'_q - d_d E'_q = E_{fd} - (d_d - c_d) E'_q = E_{fd} - E'_q \implies E'_q = E_{fd}. \quad (15)$$

So the identity (14) is exactly what makes the open-circuit field EMF equal the applied field voltage — the standard consistency check for this flux parameterisation, and the reason the coefficients are written as ratios (13) rather than as free constants. The same argument on the q-axis gives $E'_d = 0$ at open circuit when E_{fd} has no q-axis analogue.

4.2 Algebraic stator interface

The stator currents are *not* reverse-derived from power; they are solved from the subtransient EMFs behind subtransient reactance. Projecting the complex terminal voltage V onto the machine dq frame (V_d, V_q) and writing $r_d = V_d - E''_d, r_q = V_q - E''_q$,

$$\det = X''_d X''_q + R_a^2, \quad I_d = \frac{-R_a r_d - X''_q r_q}{\det}, \quad I_q = \frac{X''_d r_d - R_a r_q}{\det}, \quad (16)$$

which is the exact inverse of the subtransient stator relations $V_d = E''_d - R_a I_d + X''_q I_q, V_q = E''_q - R_a I_q - X''_d I_d$. The electrical torque (12) uses these same I_d, I_q .

4.3 The singular limit that freezes E'_d

For the machine data used here the q-axis transient open-circuit time constant is set to its singular limit $T'_{q0} = 0$ (audited round-rotor case). Then (9) is no longer differential: it degenerates to the algebraic constraint

$$0 = c_q E''_d - d_q E'_d. \quad (17)$$

For this machine $c_q = 0$, $d_q = 1$, so $E'_d \equiv 0$. The code detects $T'_{q0} = 0$ exactly and *removes* E'_d (state index 4) from the integrated set, holding it at its algebraic value 0. The SG therefore contributes **five** dynamic states $[\delta, \omega, E'_q, E''_q, E''_d]$, not six. This is a `SOURCE_DEFINED` singular limit, not a numerical convenience: it is the $T'_{q0} \rightarrow 0$ reduction of the same equations.

4.4 Breaker-open coast

When the SG breaker opens at $t = 20$ s the machine does not vanish from the state vector. Its network injection becomes zero ($I_d = I_q = 0$, $T_e = 0$) and its fluxes decay on the open-circuit branch of (8)–(11) (the armature-reaction terms drop out because $I_d = I_q = 0$); the swing (6)–(7) coasts on a frozen mechanical torque T_m . The map that reports how many states the SG integrates is context-independent: it returns $[1, 2, 3, 5, 6]$ whether the breaker is closed or open. Consequently the loss of the SG slack does *not* change the SG's own state count; the composite dimension changes only because the IBRs switch mode (Section 5).

5 Inverter-based resource: the dual-mode 20-state model

Each IBR is a single device that can present either of two controllers on a common power stage. The full state is a fixed 20-vector,

$$x_{\text{IBR}} = \left[\underbrace{i_d, i_q, V_{dc}}_{x_{\text{pl}} (3)} \mid \underbrace{x_{\text{GFL}} (8)}_{\text{indices 4–11}} \mid \underbrace{x_{\text{GFM}} (9)}_{\text{indices 12–20}} \right]^T \in \mathbb{R}^{20}. \quad (18)$$

The user's shorthand $x_{\text{IBR}} = [x_{\text{GFM}} \ x_{\text{GFL}}]$ is the same object with the shared plant block x_{pl} made explicit and the two controller blocks kept in their code order. The controller blocks are

$$x_{\text{GFL}} = [\delta_{\text{PLL}}, \xi_{\text{PLL}}, \xi_P, \xi_Q, \xi_{Id}, \xi_{Iq}, V_{d,\text{del}}, V_{q,\text{del}}], \quad (19)$$

$$x_{\text{GFM}} = [\delta_{\text{VSG}}, \omega_{\text{VSG}}, E, \xi_{Vd}, \xi_{Vq}, \xi_{Id}, \xi_{Iq}, V_{d,\text{del}}, V_{q,\text{del}}]. \quad (20)$$

Only a projection of (18) is integrated at any instant. The active-index map returns

$$\mathcal{A}_{\text{GFL}} = \{1:11\}, \quad \mathcal{A}_{\text{GFM}} = \{1:3, 12:20\}, \quad \mathcal{A}_{\text{tripped}} = \emptyset, \quad (21)$$

i.e. 11 active states in GFL, 12 in GFM, none when tripped. The shared plant block is always active; the two controller blocks are mutually exclusive.

Two operators recur below. $\Pi_I(a, b; I_{\text{max}})$ is the *circular* current limiter: if $r = \sqrt{a^2 + b^2} > I_{\text{max}}$ it scales $(a, b) \mapsto (a, b) I_{\text{max}}/r$, else it is the identity; its rejected part is the limiter residual $r_i = (a, b) - \Pi_I(a, b)$. $\mathcal{H}(e, \gamma, r_i, \text{sat})$ is the conditional anti-windup integrator: it returns 0 when the reference is clipped and the error would wind further (sat and $\gamma e r_i > 0$), and e otherwise. The instantaneous injection is $I = (i_d + j i_q) e^{j\theta} / \kappa$, $S = V \bar{I}$, $P_{\text{inv}} = \kappa \Re S$, $Q_{\text{inv}} = \kappa \Im S$, with $\kappa = S_{\text{base}}/M_{\text{base}}$ and $(v_d, v_q) = \Re, \Im (V e^{-j\theta})$.

5.1 Shared power stage (states 1–3)

The L -filter current dynamics and DC-link are common to both modes and are always active. Written one state per line in index order:

$$\frac{L}{\omega_b} \dot{i}_d = v_{d,\text{del}} - v_d - R i_d + \omega L i_q, \quad d\text{-axis filter current} \quad (22)$$

$$\frac{L}{\omega_b} \dot{i}_q = v_{q,\text{del}} - v_q - R i_q - \omega L i_d, \quad q\text{-axis filter current} \quad (23)$$

$$\dot{V}_{dc} = \frac{1}{C} \left[\frac{C}{T_{dc}} (V_{dc}^0 - V_{dc}) \right], \quad \text{DC-link voltage} \quad (24)$$

with $R = R_f$, $L = L_f$, $C = C_{dc}$, $\omega_b = 2\pi f_b$. *Proof of each row.* Rows (1)–(2) are Kirchhoff's voltage law across the series R_f, L_f filter in the rotating frame; the cross terms $\pm\omega L$ are the frame-rotation EMF, and the rotation rate ω is supplied by whichever controller is active (the PLL in GFL, the VSG swing in GFM) — this is the only place the plant equations feel the mode. Row (3) is the PROJECT_DERIVED DC-source closure: the coded current is $I_{dc} = P_{ac}/V_{dc} + (C/T_{dc})(V_{dc}^0 - V_{dc})$ and $\dot{V}_{dc} = (I_{dc} - P_{ac}/V_{dc})/C$, in which the power feed-forward P_{ac}/V_{dc} cancels exactly, leaving the first-order voltage restoration shown. This cancellation is deliberate: the unpublished DC source cannot move the sourced AC equilibrium. At steady state row (3) gives $V_{dc} = V_{dc}^0$.

5.2 GFL (states 4–11):

The auxiliary algebraic quantities are the power errors $e_P = \kappa u_1 - P_{\text{inv}}$, $e_Q = \kappa u_2 - Q_{\text{inv}}$, the raw and clipped current references

$$i_{d,\text{ref}}^{\text{raw}} = k_{pP} e_P + k_{iP} \xi_P, \quad i_{q,\text{ref}}^{\text{raw}} = -(k_{pQ} e_Q + k_{iQ} \xi_Q), \quad [i_{d,\text{ref}}, i_{q,\text{ref}}] = \Pi_I(\cdot; I_{\text{max}}), \quad (25)$$

and the inner voltage commands $v_{cd} = v_d + R i_d - \omega L i_q + k_{pI}(i_{d,\text{ref}} - i_d) + k_{iI} \xi_{Id}$, $v_{cq} = v_q + R i_q + \omega L i_d + k_{pI}(i_{q,\text{ref}} - i_q) + k_{iI} \xi_{Iq}$. The eight GFL states then evolve, one per line in index order:

$$\dot{\delta}_{\text{PLL}} = k_{p,\text{PLL}} v_q + k_{i,\text{PLL}} \xi_{\text{PLL}}, \quad \text{PLL angle; } \omega = 1 + \dot{\delta}_{\text{PLL}}/\omega_b \quad (26)$$

$$\dot{\xi}_{\text{PLL}} = v_q, \quad \text{PLL integrator} \quad (27)$$

$$\dot{\xi}_P = \mathcal{H}(e_P, k_{iP}, r_{i,d}, \text{sat}), \quad \text{active-power integrator} \quad (28)$$

$$\dot{\xi}_Q = \mathcal{H}(e_Q, -k_{iQ}, r_{i,q}, \text{sat}), \quad \text{reactive-power integrator} \quad (29)$$

$$\dot{\xi}_{Id} = i_{d,\text{ref}} - i_d, \quad d\text{-current integrator} \quad (30)$$

$$\dot{\xi}_{Iq} = i_{q,\text{ref}} - i_q, \quad q\text{-current integrator} \quad (31)$$

$$\dot{V}_{d,\text{del}} = (v_{cd} - V_{d,\text{del}})/T_d, \quad d\text{-command lag} \quad (32)$$

$$\dot{V}_{q,\text{del}} = (v_{cq} - V_{q,\text{del}})/T_d, \quad q\text{-command lag} \quad (33)$$

Proof of each state. (4)–(5) are the synchronous-reference-frame PLL: it drives the measured q -axis voltage v_q to zero, so at lock $\dot{\xi}_{\text{PLL}} = v_q = 0$ and the frame is aligned with the terminal voltage; the loop frequency is $\omega = 1 + \dot{\delta}_{\text{PLL}}/\omega_b$. (6)–(7) are the outer P/Q PIs: at equilibrium $\dot{\xi}_P = e_P = 0$ forces $P_{\text{inv}} = \kappa u_1$ and $\dot{\xi}_Q = e_Q = 0$ forces $Q_{\text{inv}} = \kappa u_2$, i.e. the IBR meets its P, Q set-points; the $\mathcal{H}$ wrapper freezes them only while the current reference is on the I_{max} circle. (8)–(9) are the inner current PIs: $\dot{\xi}_{Id} = i_{d,\text{ref}} - i_d = 0$ makes the filter current track its reference (and symmetrically on q). (10)–(11) are the first-order actuation lags: $\dot{V}_{d,\text{del}} = 0 \Rightarrow V_{d,\text{del}} = v_{cd}$, so the delayed command equals the computed command at steady state. The GFL branch is a *current source*: it needs the PLL to know the grid angle, so it cannot by itself define a voltage on a dead network.

5.3 GFM (states 12–20): voltage source, VSG,

The grid-forming branch replaces the PLL by a virtual-synchronous-generator swing and a Q – V voltage-forming loop. The voltage references are $v_{d,\text{ref}} = E$, $v_{q,\text{ref}} = 0$; the voltage errors are $e_{vd} = E - v_d$, $e_{vq} = -v_q$; the raw/clipped current references are $i_{d,\text{ref}}^{\text{raw}} = k_p v e_{vd} + k_{iV} \xi_{Vd}$, $i_{q,\text{ref}}^{\text{raw}} = k_p v e_{vq} + k_{iV} \xi_{Vq}$, $[i_{d,\text{ref}}, i_{q,\text{ref}}] = \Pi_I(\cdot; I_{\max})$; and the inner commands v_{cd}, v_{cq} are as in the GFL branch but with $\omega \rightarrow \omega_{\text{VSG}}$. The nine GFM states evolve, one per line in index order:

$$\dot{\delta}_{\text{VSG}} = \omega_b(\omega_{\text{VSG}} - 1), \quad \text{VSG angle} \quad (34)$$

$$M \dot{\omega}_{\text{VSG}} = \kappa u_1 - P_{\text{inv}} - D_v(\omega_{\text{VSG}} - 1), \quad \text{VSG swing} \quad (35)$$

$$\tau_E \dot{E} = k_Q(\kappa u_2 - Q_{\text{inv}}) - k_E(E - E_{\text{ref}}), \quad Q\text{--}V \text{ voltage state} \quad (36)$$

$$\dot{\xi}_{Vd} = \mathcal{H}(e_{vd}, k_{iV}, r_{i,d}, \text{sat}), \quad d\text{-voltage integrator} \quad (37)$$

$$\dot{\xi}_{Vq} = \mathcal{H}(e_{vq}, k_{iV}, r_{i,q}, \text{sat}), \quad q\text{-voltage integrator} \quad (38)$$

$$\dot{\xi}_{Id} = i_{d,\text{ref}} - i_d, \quad d\text{-current integrator} \quad (39)$$

$$\dot{\xi}_{Iq} = i_{q,\text{ref}} - i_q, \quad q\text{-current integrator} \quad (40)$$

$$\dot{V}_{d,\text{del}} = (v_{cd} - V_{d,\text{del}})/T_d, \quad d\text{-command lag} \quad (41)$$

$$\dot{V}_{q,\text{del}} = (v_{cq} - V_{q,\text{del}})/T_d, \quad q\text{-command lag} \quad (42)$$

Proof of each state. (12)–(13) are the VSG: the swing gives the converter its *own* angle from active-power balance, $M\dot{\omega}_{\text{VSG}} = \kappa u_1 - P_{\text{inv}} - D_v(\omega_{\text{VSG}} - 1)$, so at equilibrium $\omega_{\text{VSG}} = 1$ and $P_{\text{inv}} = \kappa u_1$; because the angle is generated internally there is *no PLL* and the branch can energise a passive network. (14) is the Q – V loop whose state E is the voltage-loop reference $v_{d,\text{ref}} = E$: at equilibrium $E = E_{\text{ref}} + (k_Q/k_E)(\kappa u_2 - Q_{\text{inv}})$, a droop between reactive set-point and formed voltage. (15)–(16) are the voltage PIs that drive $v_d \rightarrow E$, $v_q \rightarrow 0$ (so the terminal voltage aligns with the VSG angle at magnitude E); (17)–(18) the inner current PIs; (19)–(20) the same first-order command lags as GFL. The anti-windup $\mathcal{H}$ again holds the voltage integrators only while the current reference is saturated. The GFM branch is a *voltage source*: this is why the reference-loss guard promotes IBRs to GFM the instant the SG slack is lost.

5.4 Current-continuity-gated transfer

When the supervisor changes an IBR's mode, the new-mode controller states are not continued by integration; they are *re-initialised* at the current terminal operating point so that the injected current does not jump. Let the pre-switch state x^- inject $I^- = I(x^-, V)$ with $S = V\overline{I^-}$, $P = \Re S$, $Q = \Im S$. The transfer $x^- \mapsto x^+$ is

$$x_{\text{pl}}^+ = x_{\text{pl}}^- (\text{carry } i_d, i_q, V_{dc}), \quad x_{\text{ctrl}}^+ = \text{equilibrium_initialize}(V, P, Q), \quad (43)$$

with one physical carry-across of frequency: GFL→GFM sets $\omega_{\text{VSG}} := \omega_{\text{PLL}}$, and GFM→GFL sets the PLL integrator $\xi_{\text{PLL}} := \omega_b(\omega_{\text{VSG}} - 1)/k_{i,\text{PLL}}$. The transfer is then *gated*: the post-switch injection $I^+ = I(x^+, V)$ must satisfy

$$|I^+ - I^-| \leq \varepsilon_{\text{tol}} = 10^{-10}, \quad (44)$$

otherwise the step is rejected fail-closed. Equation (44) is the bumpless-transfer guarantee: the mode may change discontinuously in *parameterisation*, but the current the network sees is continuous.

6 Switched time-domain formulation

The composite system is advanced in time by *one* coupled implicit step that the plain and the hybrid drivers share. Three questions are answered in turn, because they are the three things that make this harder than integrating a single machine: (i) the differential equations are *not* the same from one device to the next, nor even for one device across a mode change (§6.1); (ii) they are nonetheless solved *together*, in a single system per step, by discretising the differential part with the implicit trapezoidal rule and appending the network law (§6.2); and (iii) that system is nonlinear, so each step is closed by a damped Newton solve (§6.3). Two refinements specific to this study — the frozen coordinates (§6.4) and the mode switch that replaces integration by a direct assignment (§6.5) — follow.

6.1 Why the device ODEs are not the same

Each device owns a different differential model, so the right-hand side is heterogeneous by construction:

- the **SG** obeys the sixth-order EMF6 equations of Section 4 (swing pair, transient and sub-transient fluxes), of which five are integrated because E'_d is frozen at the singular limit (active set $[1, 2, 3, 5, 6]$);
- an **IBR in GFL** integrates its shared power stage plus the phase-locked-loop controller block — 11 states, $\mathcal{A}_{\text{GFL}} = \{1:11\}$ of (21);
- an **IBR in GFM** integrates the same power stage plus the virtual-synchronous-machine controller (no PLL) — 12 states, $\mathcal{A}_{\text{GFM}} = \{1:3, 12:20\}$;
- a **tripped** device integrates nothing ($\mathcal{A}_{\text{tripped}} = \emptyset$).

The full state is the concatenation of these blocks, $x = [x_{\text{SG}}; x_{\text{IBR}_1}; \dots; x_{\text{IBR}_4}]$, and the right-hand side is the matching concatenation

$$\dot{x} = f(x, V) = [f_{\text{SG}}(x_{\text{SG}}, V); f_{\text{IBR}_1}(x_{\text{IBR}_1}, V; m_1); \dots], \quad (45)$$

where each block f_{IBR_i} depends on that inverter's current mode $m_i \in \{\text{GFL}, \text{GFM}, \text{tripped}\}$ and is written by the device itself. There is no single ODE that all devices share; the coupling between them is *only* through the terminal voltages V that every block reads.

6.2 One coupled DAE, one implicit trapezoidal step

The heterogeneous ODE (45) is coupled to the all-KCL network law (1) to form the composite differential–algebraic system

$$\dot{x} = f(x, V), \quad 0 = g(x, V) = Y_{\text{net}}V - I_{\text{bus}}(x, V). \quad (46)$$

It is advanced from (x_0, V_0) to (x_1, V_1) over a step h by discretising the differential rows with the implicit trapezoidal rule and carrying the algebraic rows exactly. With $f_0 = f(x_0, V_0)$, $f_1 = f(x_1, V_1)$ and the active set $\mathcal{A} \subseteq \{1, \dots, n_x\}$ (the union of the per-device active sets above), the step solves

$$\underbrace{\left[x_1 - x_0 - \frac{h}{2}(f_0 + f_1) \right]_{\mathcal{A}}}_{\text{active trapezoidal rows}} = 0, \quad \underbrace{g(x_1, V_1)}_{\text{all KCL rows}} = 0. \quad (47)$$

Because f_1 and g are evaluated at the unknown end-of-step state, this is an *implicit* (nonlinear, coupled) system, not an explicit update: the states of every device and the voltages of every bus are unknowns of one and the same system, so the machine and all four inverters are solved simultaneously and are never sub-iterated device by device. For an active state i the first block is exactly the trapezoidal rule $x_{1,i} = x_{0,i} + \frac{h}{2}(f_{0,i} + f_{1,i})$.

6.3 How the step is solved: the damped Newton system

Stacking the unknowns as $z = [x_1|_{\mathcal{A}}; V_1]$, the two blocks of (47) become a single nonlinear residual

$$r(z) = \begin{bmatrix} [x_1 - x_0 - \frac{h}{2}(f_0 + f_1)]_{\mathcal{A}} \\ g(x_1, V_1) \end{bmatrix} = 0, \quad (48)$$

which is driven to zero by a damped Newton iteration. This is the single Newton owner reused by the equilibrium solve, the trapezoidal step, and the event solve; no device supplies its own solver. One iteration is:

1. form the Jacobian $J = \partial r / \partial z$ by *forward finite differences* (column j is $[r(z + \varepsilon e_j) - r(z)] / \varepsilon$ with $\varepsilon = 3 \times 10^{-6}$); no analytic Jacobian is assembled, so a device may change its equations without hand-derived derivatives;
2. test conditioning: if $\text{rcond}(J) < \epsilon_{\text{mach}}$ the step is rejected fail-closed rather than solved through a singular system;
3. take the Newton direction $\Delta z = -(J \backslash r)$ with the audited MATLAB backslash — the only linear solve in the loop;
4. damp it with a backtracking line search: try $z + \alpha \Delta z$ at $\alpha = 1$ and halve α (up to 20 times) until $\|r\|_{\infty}$ strictly decreases; if no α decreases the residual the step fails closed at the last accepted iterate.

The iteration is declared converged when $\|r(z)\|_{\infty} < \text{tol} = 10^{-8}$, and at most 50 iterations are taken. Nothing in the loop is tuned to the case: the tolerance, the finite-difference increment, the conditioning gate and the line-search contract are fixed constants of the solver.

6.4 Frozen coordinates: the Π -projection

Every index *not* in $\mathcal{A}$ — the frozen set $\mathcal{F} = \{1, \dots, n_x\} \setminus \mathcal{A}$ — is removed from the Newton system and held exactly:

$$x_{1,i} = x_{0,i} \quad \text{for all } i \in \mathcal{F}. \quad (49)$$

The solver begins from $x_1 \leftarrow x_0$ and never writes a frozen row, and a drift check re-verifies $x_1|_{\mathcal{F}} = x_0|_{\mathcal{F}}$ after the step; any change of a frozen coordinate is a hard error. Two distinct things live in $\mathcal{F}$: the singular-limit flux E'_d of Section 4.3 (frozen for the whole run), and the entire controller block that a given IBR is *not* currently using (frozen for as long as that mode is inactive). The map Π that selects $\mathcal{A}$ is the same runtime authority (`ts_dynamic_state_indices`) at every one of its call sites, so the figure of $n_x(t)$ below cannot disagree with what the integrator actually solved.

6.5 The switch: from integration to assignment

A mode change is landed *exactly* on a step boundary (the step is subdivided so the event time is a grid point). At that boundary the state is not advanced by (47); it is replaced by the discrete transfer of Section 5.4:

$$x_1 = \mathcal{T}(x_0, V) \quad (\text{a re-initialisation}), \text{ not } x_1 = x_0 + \frac{h}{2}(f_0 + f_1). \quad (50)$$

At the mode boundary the controller coordinates jump to the transferred values $\mathcal{T}(x_0, V)$ chosen to satisfy the current-continuity gate (44), and integration resumes on the *new* active set on the very next step. Two important consequences follow for this study:

- The **SG trip** is *not* such an assignment. The SG keeps the same active set [1, 2, 3, 5, 6]; its trajectory is continuous across $t = 20$ s (the coast of Section 4). Only the right-hand side branch changes (connected stator $\rightarrow$ open circuit).
- The **IBR GFL $\rightarrow$ GFM** changes at $t = 20$ s *are* such assignments: each IBR's active dimension steps $11 \rightarrow 12$ and its controller states are re-initialised by (43). The composite dynamic dimension therefore changes because of the inverters, not the generator.

7 Power-flow results

Table 1 is the solved SG-online power flow (the reference operating point from which V_i^* in (2) is taken); Table 2 the corresponding branch flows and losses. The reference bus supplies $P_g = 1.3319$ p.u., $Q_g = -0.1957$ p.u.; the four IBR buses (2, 3, 6, 8) inject at their scheduled set-points while every load bus meets its P_L, Q_L .

Table 1: Solved bus power flow at the SG-online operating point (per unit on the system base; angles in degrees).

Bus	Type	V	θ (deg)	P_g	Q_g	P_L	Q_L	Q_{min}	Q_{max}	P_{max}
1	REF	1.0000	0.0000	1.3319	-0.1957	0.0000	0.0000	0.00	0.10	3.32
2	GFL	0.9914	-3.2585	0.3172	0.1767	0.2170	0.1270	-0.40	0.50	1.40
3	GFL	0.9667	-8.6921	0.2884	0.1607	0.9420	0.1900	0.00	0.40	1.00
4	PQ	0.9832	-6.7064	0.0000	0.0000	0.4780	-0.0390	–	–	–
5	PQ	0.9858	-5.5221	0.0000	0.0000	0.0760	0.0160	–	–	–
6	GFL	1.0575	-6.7136	0.4443	0.2476	0.1120	0.0750	-0.06	0.24	1.00
7	PQ	1.0253	-6.9205	0.0000	0.0000	0.0000	0.0000	–	–	–
8	GFL	1.0494	-4.4090	0.2677	0.1491	0.0000	0.0000	-0.06	0.24	1.00
9	PQ	1.0219	-8.6416	0.0000	0.0000	0.2950	0.1660	–	–	–
10	PQ	1.0202	-8.5886	0.0000	0.0000	0.0900	0.0580	–	–	–
11	PQ	1.0347	-7.7782	0.0000	0.0000	0.0350	0.0180	–	–	–
12	PQ	1.0411	-7.6862	0.0000	0.0000	0.0610	0.0160	–	–	–
13	PQ	1.0342	-7.8345	0.0000	0.0000	0.1350	0.0580	–	–	–
14	PQ	1.0087	-9.3374	0.0000	0.0000	0.1490	0.0500	–	–	–

Table 2: Branch power flows and losses at the SG-online operating point.

Line	From	To	P_{from}	Q_{from}	P_{loss}	Q_{loss}
1	1	2	0.910918	-0.153112	0.016392	-0.002303
2	1	5	0.420987	-0.042553	0.009593	-0.008907
3	2	3	0.466697	0.013221	0.010470	0.002115
4	2	4	0.316733	-0.064476	0.006065	-0.014738
5	2	5	0.211276	-0.049810	0.002649	-0.025732
6	3	4	-0.197333	-0.018164	0.002803	-0.005014
7	4	5	-0.448842	0.084724	0.002882	0.009089
8	4	7	0.018419	-0.096315	0.000000	0.001990
9	4	9	0.062955	-0.012297	0.000000	0.002223
10	5	6	0.092297	0.001911	0.000000	0.001920
11	6	11	0.130821	0.059907	0.001758	0.003682
12	6	12	0.086088	0.027130	0.000895	0.001864
13	6	13	0.207667	0.085524	0.002983	0.005875
14	7	8	-0.267650	-0.134127	0.000000	0.015018
15	7	9	0.286069	0.035823	0.000000	0.008698
16	9	10	-0.003258	0.021664	0.000015	0.000039
17	9	14	0.057282	0.023368	0.000466	0.000991
18	10	11	-0.093273	-0.036375	0.000790	0.001849
19	12	13	0.024192	0.009266	0.000137	0.000124
20	13	14	0.093739	0.030791	0.001556	0.003168

8 Small-signal spectrum

The modal table below lists the small-signal eigenvalues at a representative post-disturbance configuration (one IBR grid-forming, three grid-following): index, eigenvalue λ , modal frequency f , damping ratio ζ , and the single dominant participating state. Conjugate pairs are printed once. The table is a presentation-only column projection of the generated modal table — only the trailing participation entries are dropped; no eigenvalue, frequency or damping ratio is recomputed, reordered or reformatted, so the numbers are bit-identical to the full table.

No.	Eigenvalue λ (s^{-1})	f (Hz)	ζ	Dominant state
1	$-3881.470073 + j1061.916124$	169.009200	0.964553	IBR6: i_d (0.131)
2	-159.046981	0	1.000000	IBR8: E^{GFM} (0.992)
3	$-156.112829 + j816.094065$	129.885404	0.187886	IBR6: i_d (0.305)
4	$-140.724047 + j1033.942353$	164.557036	0.134861	IBR8: i_q (0.344)
5	$-106.207976 + j615.792052$	98.006349	0.169964	IBR2: i_d (0.248)
6	-17.922231	0	1.000000	IBR8: ω_{VSG}^{GFM} (0.909)
7	$-13.345138 + j1.527324$	0.243081	0.993514	IBR2: ξ_{Id}^{GFL} (0.261)
8	$-13.151005 + j2.154763$	0.342941	0.986841	IBR6: ξ_{Id}^{GFL} (0.281)
9	$-12.633262 + j3.598177$	0.572668	0.961751	IBR8: ξ_{Id}^{GFL} (0.227)
10	$-11.446947 + j14.627753$	2.328079	0.616280	IBR8: $V_{d,del}^{GFM}$ (0.155)
11	-10.000000	0	1.000000	IBR2: V_{dc} (1.000)
12	-10.000000	0	1.000000	IBR3: V_{dc} (1.000)
13	-10.000000	0	1.000000	IBR6: V_{dc} (1.000)
14	-10.000000	0	1.000000	IBR8: V_{dc} (1.000)

No.	Eigenvalue λ (s^{-1})	f (Hz)	ζ	Dominant state
15	$-6.988836 + j12.831357$	2.042174	0.478320	IBR8: $V_{q,del}^{GFM}$ (0.155)
16	$-6.293824 + j109.583760$	17.440797	0.057339	IBR2: $V_{d,del}^{GFL}$ (0.224)
17	$-2.508684 + j84.237836$	13.406868	0.029768	IBR6: $V_{d,del}^{GFL}$ (0.256)
18	-2.048337	0	1.000000	IBR6: ξ_P^{GFL} (0.289)
19	$-1.643720 + j47.532682$	7.565061	0.034560	IBR8: $V_{d,del}^{GFM}$ (0.219)
20	-1.509781	0	1.000000	IBR6: ξ_Q^{GFL} (0.321)
21	$-1.503650 + j0.674893$	0.107413	0.912318	IBR8: ξ_{Vq}^{GFM} (0.286)
22	-1.443329	0	1.000000	IBR3: ξ_P^{GFL} (0.299)
23	-1.407957	0	1.000000	IBR2: ξ_Q^{GFL} (0.343)
24	-1.390336	0	1.000000	IBR3: ξ_Q^{GFL} (0.376)
25	-0.746742	0	1.000000	IBR8: ξ_{Vd}^{GFM} (0.624)
26	$-0.619437 + j2.216829$	0.352819	0.269116	IBR6: δ_{PLL}^{GFL} (0.278)
27	$-0.605533 + j2.186111$	0.347930	0.266940	IBR3: δ_{PLL}^{GFL} (0.305)
28	$-0.175750 + j2.159250$	0.343655	0.081126	IBR3: δ_{PLL}^{GFL} (0.136)

The spectrum is entirely in the left half-plane at this configuration. This is a small-signal statement about one linearisation point; it is not, and must not be read as, a stability certificate for the full switched trajectory.

9 Time-Domain Simulation

Figure 2 shows the per-IBR active and reactive power over the full preserved horizon 0–200 s. Two strips above the power panels give the supervisory state: the top strip is each IBR’s grid-forming/grid-following mode (all four step to GFM together when the SG is lost), and the second strip is the *reference owner* — the single device that holds the island angle reference at each instant. Mode and reference ownership are distinct contracts: several IBRs can be grid-forming at once, but only one of them owns the reference. Here the SG owns it up to $t = 20$ s and again after the reclose; in between, IBR₁ (bus 2) is the reference leader (IBR₄ holds it only for the first fraction of a second before the supervisor settles on IBR₁). The four traces are the four IBRs at buses 2, 3, 6 and 8. Every sample is a raw accepted value of the preserved 200-s trajectory; the 0–200 s range is the full preserved horizon (no display truncation).

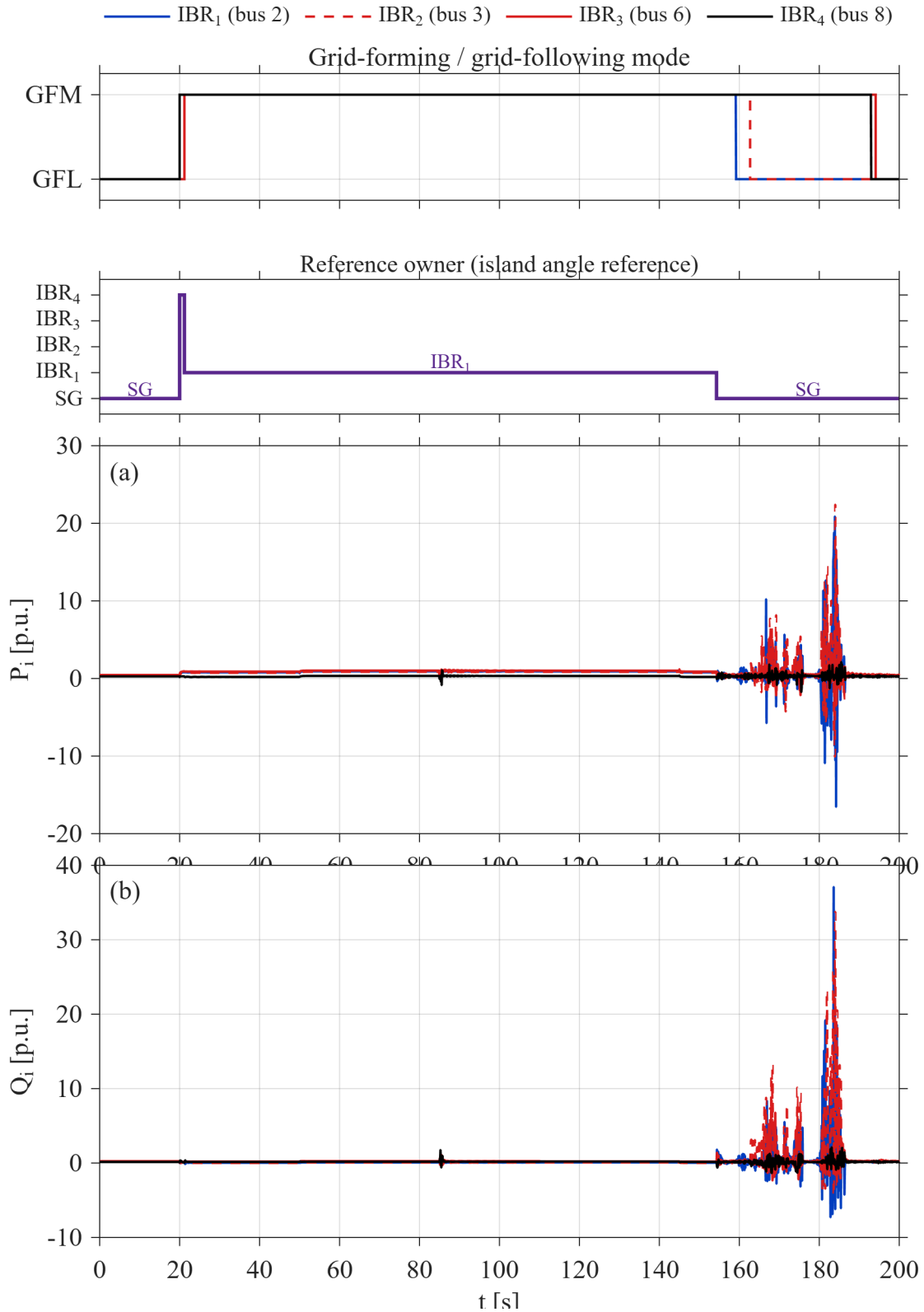


Figure 2: Top strip: grid-forming/grid-following mode of each IBR (GFL/GFM step, four traces overlaid, no display offset). Second strip: reference owner (which device holds the island angle reference — SG, then IBR₁, then SG again). (a) Per-IBR active power P_i p.u.. (b) Per-IBR reactive power Q_i p.u.. Raw accepted samples of the preserved 200-s trajectory; $t \in [0, 200]$ s full window.

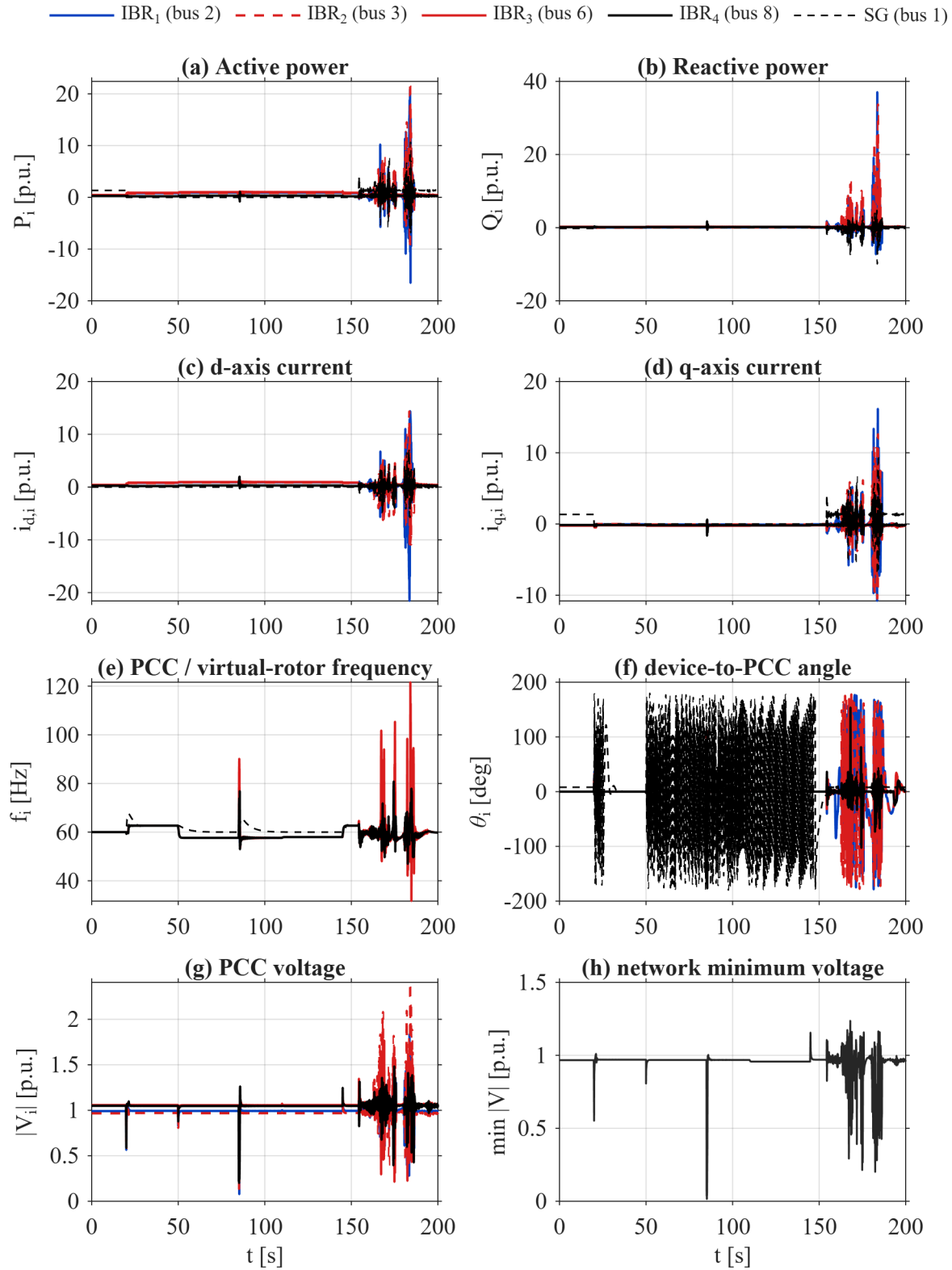


Figure 3: Eight-panel electrical response (raw accepted samples, preserved 200-s trajectory, $t \in [0, 200]$ s full window). (a) active power P_i ; (b) reactive power Q_i ; (c) d -axis current; (d) q -axis current; (e) PCC / virtual-rotor frequency; (f) device-to-PCC angle; (g) PCC voltage; (h) network minimum voltage. IBR₁ blue, IBR₂ red dashed, IBR₃ red, IBR₄ black; SG black dashed.

10 Limitations

- **No operational-readiness claim.** That the 200-s trajectory is an accepted, converged solution of the composite DAE means only that the numerics are self-consistent. It is not evidence of grid-code compliance, protection coordination or hardware realisability.
- **Transient extrema are extrema, not ratings.** The accepted trajectory reaches a peak network voltage of 2.406517 p.u. and a COI frequency band of 55.390979–84.329308 Hz during the fault and switching window. These are honest excursions of the model under a severe chronology, not steady operating values, and they underline that the study is a numerical exercise, not a design sign-off.
- **One linearisation.** The spectrum of Section 8 is a single operating-point analysis; it does not bound the switched system.
- **Reporting reconstruction.** J_V, J_f in the report are reconstructed from the accepted COI frequency and healthy per-bus PF voltage for exposition; the run-time supervisor uses its own in-loop signals. The two agree in intent, not necessarily sample-for-sample.
- **Figure horizon.** Equations are those of the 250-s code; the plotted curves are the preserved 200-s run. When the 250-s trajectory is finalised the figures and the frozen run scalars must be regenerated together.